

Manipulator jacobians

A matrix relating joint param time der
with end effector dynamics

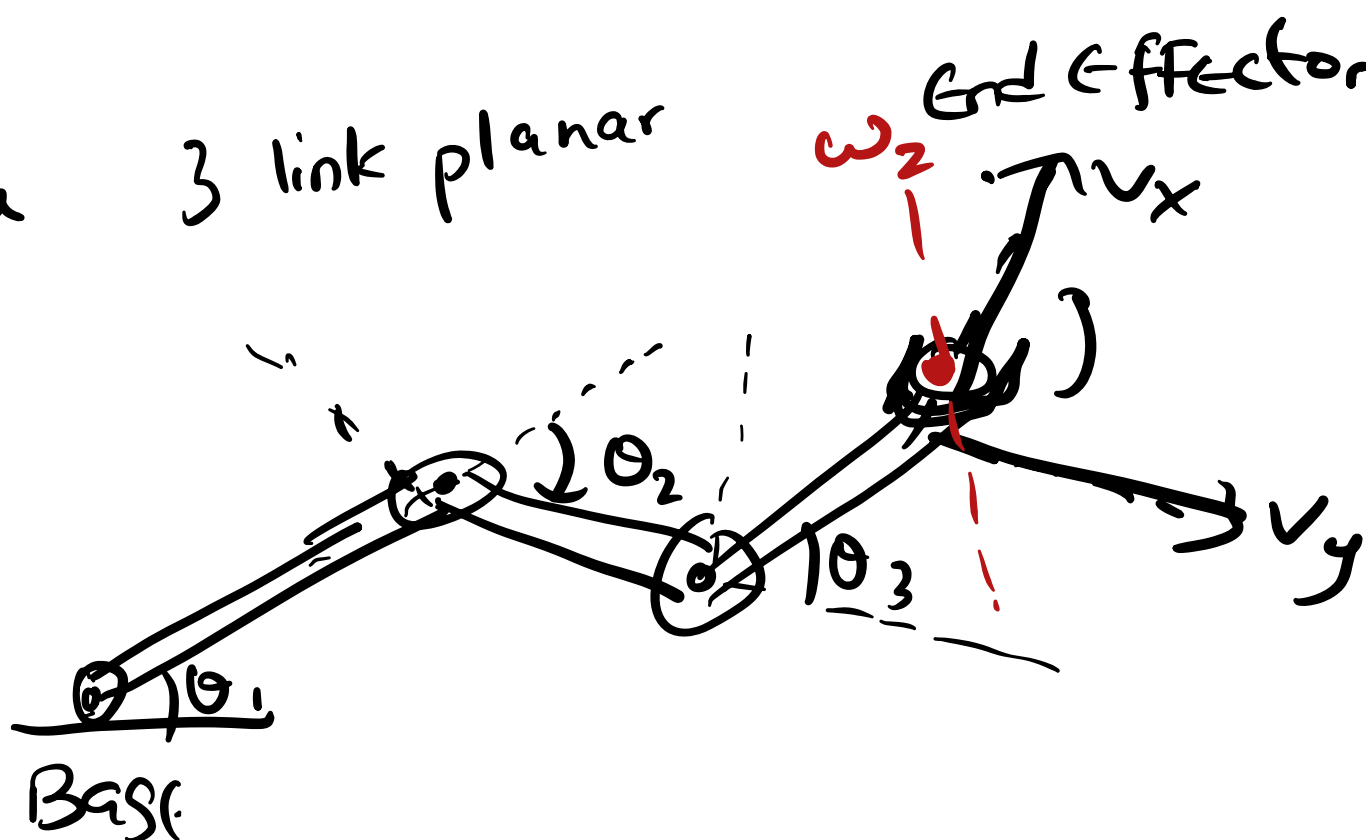
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Twist

$$v = \begin{bmatrix} v_x \\ v_y \\ v_z \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

(of the
end
effector)

$$\dot{q} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

For a 3 link planar



Twist $v = \begin{bmatrix} v_x \\ v_y \\ 0 \\ 0 \\ 0 \\ \omega_z \end{bmatrix}$ for a planar arm

chain rule
 $v_x = \dot{x} = \frac{dx}{dt}$ From Fk we know x is fⁿ of $\theta_1, \theta_2, \theta_3$

$$x(\theta_1, \theta_2, \theta_3) = \frac{\partial x}{\partial \theta_1} \dot{\theta}_1 + \frac{\partial x}{\partial \theta_2} \dot{\theta}_2 + \frac{\partial x}{\partial \theta_3} \dot{\theta}_3$$

[as $\frac{\partial x}{\partial \theta_1} \frac{\partial \theta_1}{\partial t} = \frac{\partial x}{\partial \theta_1} \dot{\theta}_1$]

$$\dot{x} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \end{bmatrix}_{1 \times 3} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} \end{bmatrix}_{1 \times 1}$$

Similarly for y

$$\dot{y} = \begin{bmatrix} \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

$$\omega_z = \dot{\phi}_z$$

where ϕ_z is end effector orientation

$$\phi_z = \theta_1 + \theta_2 + \theta_3$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\phi}_z \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

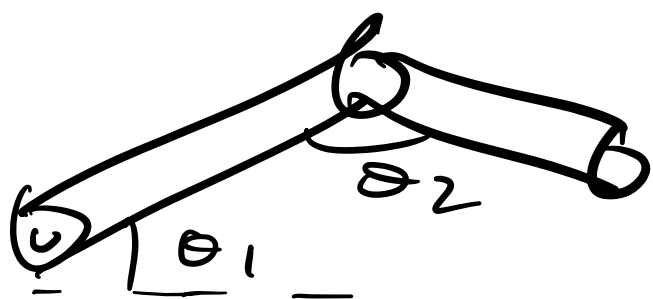
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"Jacobian matrix"

"Just chain rule for 3 different variables written in the form of a matrix!!"

Now that we know where the jacobian came from. How do we use it?

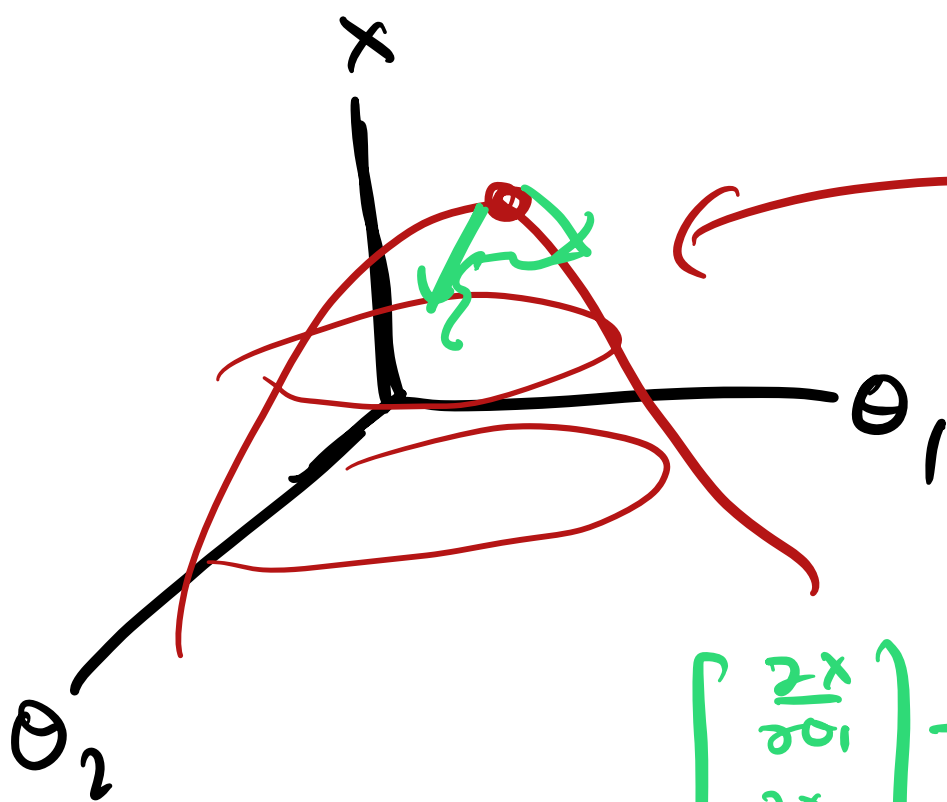
Visual meaning

consider a 2 link manipulator



$x(\theta_1, \theta_2)$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$



FK gives you shape of the function

$\begin{bmatrix} \frac{\partial x}{\partial \theta_1} \\ \frac{\partial x}{\partial \theta_2} \end{bmatrix}$ gives you Gradient

(must have studied in maths-2 after minor)

Similarly for 3 variables
the jacobian rows represent
gradient

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \nabla_q x^T \\ \nabla_q y^T \\ 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

very important to keep in mind

The objective of inverse kinematics

$$\begin{bmatrix} x(q) \\ y(q) \\ \phi(q) \end{bmatrix} \rightarrow \begin{bmatrix} x_d \\ y_d \\ \phi_d \end{bmatrix}$$

our end effector position desired end effector position

$$\overline{X}(q) \rightarrow \overline{X}_d$$

Alternative

$$e = \overline{X}(q) - \overline{X}_d$$

$$\begin{bmatrix} x(q) - x_d \\ y(q) - y_d \\ \phi(q) - \phi_d \end{bmatrix} = \begin{bmatrix} e_x(q) \\ e_y(q) \\ e_\phi(q) \end{bmatrix}$$

want magnitude of error to be 0

$$E = \frac{1}{2} \|\overline{X}_d - \overline{X}_c\|^2$$

$$E = \frac{1}{2} \left(\underbrace{e_x^2}_{(x(q)-x_d)^2} + \underbrace{e_y^2}_{(y(q)-y_d)^2} + \underbrace{e_\phi^2}_{(\phi(q)-\phi_d)^2} \right)$$

goal is to minimise this error

$x_c, y_c, \phi_c \rightarrow$ all functions of $(\theta_1, \theta_2, \theta_3)$

Thus
$$E(q) = \frac{1}{2} \|\overline{X}_d - \overline{X}_c(q)\|^2$$

we want to find a q
to minimise $E(q)$

Gradient descent

Find
 $\nabla_q \bar{E}(q)$

$$q_{\text{next}} = q_{\text{current}} - \nabla_q \bar{E}(q)$$

↓
(helps you reach minima)
Why: self study gradient descent

$$E(q) = \frac{1}{2} (e_x^2 + e_y^2 + e_\phi^2)$$

$$\frac{\partial E}{\partial \theta_1} = \frac{1}{2} \times 2 \left[e_x \frac{\partial e_x}{\partial \theta_1} + e_y \frac{\partial e_y}{\partial \theta_1} + e_\phi \frac{\partial e_\phi}{\partial \theta_1} \right]$$

$$e_x = \underset{\substack{\downarrow \\ \text{constant}}}{X(q)} - x_d$$

$$\frac{\partial e_x}{\partial \theta_1} = \underset{\substack{\downarrow}}{\frac{\partial x}{\partial \theta_1}} - 0$$

$$\left[\text{So } \frac{\partial e_x}{\partial \theta_1} = \frac{\partial x}{\partial \theta_1}, \frac{\partial e_y}{\partial \theta_1} = \frac{\partial y}{\partial \theta_1}, \frac{\partial e_\phi}{\partial \theta_1} = \frac{\partial \phi}{\partial \theta_1} \right]$$

$$\frac{\partial E}{\partial \theta_1} = e_x \frac{\partial x}{\partial \theta_1} + e_y \frac{\partial y}{\partial \theta_1} + e_z \frac{\partial z}{\partial \theta_1}$$

$$\frac{\partial E}{\partial \theta_1} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial y}{\partial \theta_1} & \frac{\partial \phi}{\partial \theta_1} \end{bmatrix} \begin{bmatrix} e_x \\ e_y \\ e_\phi \end{bmatrix}$$

Similarly

$$\frac{\partial E}{\partial \theta_2} = \begin{bmatrix} \frac{\partial x}{\partial \theta_2} & \frac{\partial y}{\partial \theta_2} & \frac{\partial \phi}{\partial \theta_2} \end{bmatrix} \begin{bmatrix} e_x \\ e_y \\ e_\phi \end{bmatrix}$$

$$\frac{\partial E}{\partial \theta_3} = \begin{bmatrix} \frac{\partial x}{\partial \theta_3} & \frac{\partial y}{\partial \theta_3} & \frac{\partial \phi}{\partial \theta_3} \end{bmatrix} \begin{bmatrix} e_x \\ e_y \\ e_\phi \end{bmatrix}$$

$$\nabla_q E = \begin{bmatrix} \frac{\partial E}{\partial \theta_1} \\ \frac{\partial E}{\partial \theta_2} \\ \frac{\partial E}{\partial \theta_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial y}{\partial \theta_1} & \frac{\partial \phi}{\partial \theta_1} \\ \frac{\partial x}{\partial \theta_2} & \frac{\partial y}{\partial \theta_2} & \frac{\partial \phi}{\partial \theta_2} \\ \frac{\partial x}{\partial \theta_3} & \frac{\partial y}{\partial \theta_3} & \frac{\partial \phi}{\partial \theta_3} \end{bmatrix} \begin{bmatrix} e_x \\ e_y \\ e_\phi \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial y}{\partial \theta_1} & 1 \\ \frac{\partial x}{\partial \theta_2} & \frac{\partial y}{\partial \theta_2} & 1 \\ \frac{\partial x}{\partial \theta_3} & \frac{\partial y}{\partial \theta_3} & 1 \end{bmatrix}$$

$$\phi = \theta_1 + \theta_2 + \theta_3$$

$$\frac{\partial \phi}{\partial \theta_1} = \frac{\partial \phi}{\partial \theta_2} = \frac{\partial \phi}{\partial \theta_3} = 1$$

Yeh rishta kya kehata hai?

$$\begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \\ 1 & 1 & 1 \end{bmatrix}$$

↓
"Jacobian matrix"

$$\nabla_q E = J^T \bar{e}$$

Thus gradient descent will look like $\Rightarrow$

$$q_{\text{next step}} = q_{\text{current step}} - J^T e$$

Thus using only the fwd kinematics relationships we can get J matrix

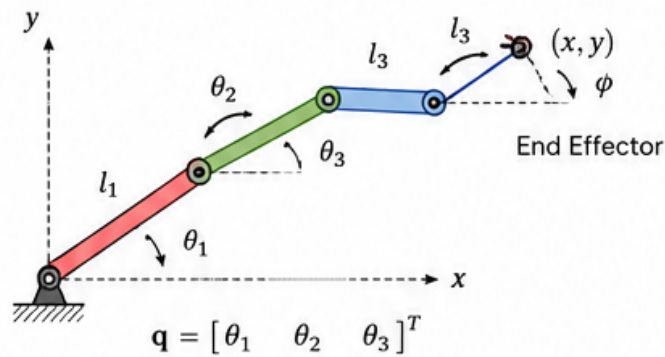
From J matrix we can use gradient descent to find the $q \equiv [\theta_1, \theta_2, \theta_3]$ to minimise error and reach

$\begin{bmatrix} x_d \\ y_d \\ \phi_d \end{bmatrix}$ $\leftarrow$ desired end effector pos'n and orientation

INVERSE KINEMATICS WITH GRADIENT DESCENT

DERIVATION FROM FIRST PRINCIPLES FOR A 3-LINK PLANAR MANIPULATOR (x, y, ϕ)

1 3-LINK PLANAR MANIPULATOR



2 FORWARD KINEMATICS

$$\begin{aligned} \theta_{12} &= \theta_1 + \theta_2 \\ \theta_{123} &= \theta_1 + \theta_2 + \theta_3 \\ \mathbf{x} &= l_1 \cos \theta_1 + l_2 \cos \theta_{12} + l_3 \cos \theta_{123} \\ y &= l_1 \sin \theta_1 + l_2 \sin \theta_{12} + l_3 \sin \theta_{123} \\ \phi &= \theta_1 + \theta_2 + \theta_3 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \mathbf{x}(\mathbf{q}) = \begin{bmatrix} x(q) \\ y(q) \\ \phi(q) \end{bmatrix} \in \mathbb{R}^3$$

3 DESIRED POSE AND ERROR

Desired pose

$$\mathbf{x}_d = \begin{bmatrix} x_d \\ y_d \\ \phi_d \end{bmatrix}$$

Error

$$\mathbf{e} = \mathbf{x}(\mathbf{q}) - \mathbf{x}_d = \begin{bmatrix} e_x \\ e_y \\ e_\phi \end{bmatrix}$$

where

$$e_x = x - x_d, \quad e_y = y - y_d, \quad e_\phi = \phi - \phi_d$$

4 OBJECTIVE (COST) FUNCTION

We want $\mathbf{e} = \mathbf{0}$. Minimize the squared error:

$$E(\mathbf{q}) = \frac{1}{2} (\mathbf{e}_x^2 + \mathbf{e}_y^2 + \mathbf{e}_\phi^2)$$

$$E : \mathbb{R}^3 \rightarrow \mathbb{R}$$

5 GRADIENT OF E : DIFFERENTIATE COMPONENT-WISE (CHAIN RULE)

Derivative w.r.t. θ_1

$$\frac{\partial E}{\partial \theta_1} = e_x \frac{\partial x}{\partial \theta_1} + e_y \frac{\partial y}{\partial \theta_1} + e_\phi \frac{\partial \phi}{\partial \theta_1}$$

$$\begin{aligned} \frac{\partial x}{\partial \theta_1} &= -l_1 \sin \theta_1 - l_2 \sin \theta_{12} - l_3 \sin \theta_{123} \\ \frac{\partial y}{\partial \theta_1} &= l_1 \cos \theta_1 + l_2 \cos \theta_{12} + l_3 \cos \theta_{123} \\ \frac{\partial \phi}{\partial \theta_1} &= 1 \end{aligned}$$



$$\begin{aligned} \frac{\partial E}{\partial \theta_1} &= e_x (-l_1 \sin \theta_1 - l_2 \sin \theta_{12} - l_3 \sin \theta_{123}) \\ &+ e_y (l_1 \cos \theta_1 + l_2 \cos \theta_{12} + l_3 \cos \theta_{123}) \\ &+ e_\phi \end{aligned}$$

Derivative w.r.t. θ_2

$$\frac{\partial E}{\partial \theta_2} = e_x \frac{\partial x}{\partial \theta_2} + e_y \frac{\partial y}{\partial \theta_2} + e_\phi \frac{\partial \phi}{\partial \theta_2}$$

$$\begin{aligned} \frac{\partial x}{\partial \theta_2} &= -l_2 \sin \theta_{12} - l_3 \sin \theta_{123} \\ \frac{\partial y}{\partial \theta_2} &= l_2 \cos \theta_{12} + l_3 \cos \theta_{123} \\ \frac{\partial \phi}{\partial \theta_2} &= 1 \end{aligned}$$



$$\begin{aligned} \frac{\partial E}{\partial \theta_2} &= e_x (-l_2 \sin \theta_{12} - l_3 \sin \theta_{123}) \\ &+ e_y (l_2 \cos \theta_{12} + l_3 \cos \theta_{123}) \\ &+ e_\phi \end{aligned}$$

Derivative w.r.t. θ_3

$$\frac{\partial E}{\partial \theta_3} = e_x \frac{\partial x}{\partial \theta_3} + e_y \frac{\partial y}{\partial \theta_3} + e_\phi \frac{\partial \phi}{\partial \theta_3}$$

$$\begin{aligned} \frac{\partial x}{\partial \theta_3} &= -l_3 \sin \theta_{123} \\ \frac{\partial y}{\partial \theta_3} &= l_3 \cos \theta_{123} \\ \frac{\partial \phi}{\partial \theta_3} &= 1 \end{aligned}$$



$$\begin{aligned} \frac{\partial E}{\partial \theta_3} &= e_x (-l_3 \sin \theta_{123}) \\ &+ e_y (l_3 \cos \theta_{123}) \\ &+ e_\phi \end{aligned}$$

6 GRADIENT VECTOR

Assemble the partial derivatives:

$$\nabla E(\mathbf{q}) = \begin{bmatrix} \frac{\partial E}{\partial \theta_1} \\ \frac{\partial E}{\partial \theta_2} \\ \frac{\partial E}{\partial \theta_3} \end{bmatrix}$$

Compact form:

Each component is a dot product between the error vector and a column of the Jacobian.

7 THE JACOBIAN

Jacobian (3×3):

$$\mathbf{J}(\mathbf{q}) = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} & \frac{\partial x}{\partial \theta_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} & \frac{\partial y}{\partial \theta_3} \\ \frac{\partial \phi}{\partial \theta_1} & \frac{\partial \phi}{\partial \theta_2} & \frac{\partial \phi}{\partial \theta_3} \end{bmatrix}$$

Its columns are:

$$\mathbf{j}_i = \begin{bmatrix} \frac{\partial x}{\partial \theta_i} \\ \frac{\partial y}{\partial \theta_i} \\ \frac{\partial \phi}{\partial \theta_i} \end{bmatrix} \quad i = 1, 2, 3$$

$$\text{Therefore,} \\ \nabla E = \mathbf{J}^T \mathbf{e}$$

(Because $\partial E / \partial \theta_i = \mathbf{e}^T \mathbf{j}_i$)

8 GRADIENT DESCENT UPDATE

Move opposite to the gradient (steepest descent):

$$\mathbf{q}_{k+1} = \mathbf{q}_k - \alpha \nabla E(\mathbf{q}_k)$$

Substitute $\nabla E = \mathbf{J}^T \mathbf{e}$:

$$\mathbf{q}_{k+1} = \mathbf{q}_k - \alpha \mathbf{J}^T(\mathbf{q}_k) \mathbf{e}(\mathbf{q}_k)$$

where $\alpha > 0$ is the step size (learning rate).

Interpretation:

The error vector $\mathbf{e}$ in task space is projected into joint space by $\mathbf{J}^T$, giving the direction that most rapidly reduces the squared pose error.

9 ALGORITHM (JACOBIAN TRANSPOSE IK)

1. Choose an initial guess $\mathbf{q}_0$.
2. Compute forward kinematics $\mathbf{x}(\mathbf{q}_k) = [x, y, \phi]^T$.
3. Compute error $\mathbf{e}_k = \mathbf{x}(\mathbf{q}_k) - \mathbf{x}_d$.
4. Compute Jacobian $\mathbf{J}(\mathbf{q}_k)$.
5. Update: $\mathbf{q}_{k+1} = \mathbf{q}_k - \alpha \mathbf{J}(\mathbf{q}_k)^T \mathbf{e}_k$.
6. Repeat until $\|\mathbf{e}_k\| < \epsilon$ (tolerance).

Notes

- α should be small enough for stability.
- Close to the solution, the method converges.
- $\mathbf{J}^T$ is simple and robust, but may converge slowly compared to pseudoinverse methods.

KEY RESULT

Starting from the squared pose error cost and applying the chain rule, we obtained

$$\nabla E = \mathbf{J}^T \mathbf{e}$$

and the gradient descent update

$$\mathbf{q}_{k+1} = \mathbf{q}_k - \alpha \mathbf{J}^T \mathbf{e}$$

This is the Jacobian Transpose (Gradient Descent) method for inverse kinematics.

x, y : position of end effector

ϕ : orientation (heading)

$\mathbf{e}$: pose error

$\mathbf{J}$: Jacobian (maps joint rates $\rightarrow$ end-effector twist)

$\mathbf{J}^T \mathbf{e}$: maps Cartesian error $\rightarrow$ joint update

Gpt
ahh
Summary

